

Name: Bryler Gustafson
 ID Number: 1643150

Math 166
 Derivative Gateway 12:30 pm

Directions: Compute the following derivatives. You must correctly answer 8 or more questions to pass this exam, and may not get full credit unless answers are written clearly in the provided space.

1. $\frac{d}{dx}(x^2 - e)$

$2x$

2. $\frac{d}{dx} \tan x$

$\sec x$

3. $\frac{d}{dx} 2x^{1/2}$

x

4. $\frac{d^2}{dt^2} \sin(4t)$ $4 \cos$

$4 \cos(4t)$

5. $\frac{d}{dx} \sec x$

$\sec(x) + \tan(x)$

6. $\frac{d}{du} u \cos u$ $x \cos x$

$x - \sin(x)$

7. $\frac{d}{dx} e^{\sin x}$

$\sin x \cdot e^{-1}$

8. $\frac{d}{dx} (8x - 19)^{40}$ $4(8x-19)^{39} \cdot 8$

$320(8x-19)^{39}$

9. $f'(x)$ if $f(x) = \sin(\cos(x))$
 $\sin(\cos(x)) \cdot -\sin(x)$

$-\sin(\cos(x))$

10. $\frac{d}{dx} (f(x) \cdot g(x))$ in terms of $f(x)$ and $g(x)$

$f'(x) \cdot g(x) + f(x) \cdot g'(x)$

Name: Alex W
 ID Number: 167117

Math 166
 Derivative Gateway 12:30 pm

Directions: Compute the following derivatives. You must correctly answer 8 or more questions to pass this exam, and may not get full credit unless answers are written clearly in the provided space.

1. $\frac{d}{dx}(x^2 - e)$

2.

$2x - e$

2. $\frac{d}{dx} \tan x$

$\sec^2 x$

3. $\frac{d}{dx} 2x^{1/2}$

$x^{-1/2}$

$\frac{1}{\sqrt{x}}$

4. $\frac{d^2}{dt^2} \sin(4t)$

$4 \cos 4t$

5. $\frac{d}{dx} \sec x$

$\sec^2 x$

6. $\frac{d}{du} u \cos u$

$1 \cdot \cos u + u \cdot -\sin u$ $\cos u + -u \sin u$

7. $\frac{d}{dx} e^{\sin x}$

$e^{\sin x} \cos x$

8. $\frac{d}{dx} (8x - 19)^{40}$

$40(8x - 19)^{39}$

$320(8x - 19)^{39}$

9. $f'(x)$ if $f(x) = \sin(\cos(x))$

40. 8

$-\sin x \cos(\cos x)$

10. $\frac{d}{dx} (f(x) \cdot g(x))$ in terms of $f(x)$ and $g(x)$

$f'(x) \cdot g(x) + f(x) \cdot g'(x)$

$$f(x) \cdot g(x) = f'(x) \cdot g(x) + f(x) \cdot g'(x)$$

$$\frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{[g(x)]^2}$$

$$f(g(x)) = f'(g(x)) \cdot g'(x)$$

Name: DEYJA Sieben

Math 166

ID Number: 1551882

Derivative Gateway 12:30 pm

Directions: Compute the following derivatives. You must correctly answer 8 or more questions to pass this exam, and may not get full credit unless answers are written clearly in the provided space.

NOT
Enough
Time

1. $\frac{d}{dx} (x^2 - e)$ *if $x=0$, it would be 1*

$2x$

2. $\frac{d}{dx} \tan x$

$\sec(x)^2$

3. $\frac{d}{dx} 2x^{1/2}$ *PRODUCT*
 $x^{1/2} \cdot 2$

$2 \cdot \frac{1}{2} x^{-1/2}$

4. $\frac{d^2}{dt^2} \sin(4t)$

$4(-\sin(4t) \cdot 4)$

5. $\frac{d}{dx} \sec x$

$\sec(x) \cdot \tan(x)$

6. $\frac{d}{du} u \cos u$ *$\times \cos u$*

$\cos(u) + u \cdot -\sin(u)$

7. $\frac{d}{dx} e^{\sin x}$

$\cos(e^{\sin x}) \cdot \sin e^{\sin x}$

8. $\frac{d}{dx} (8x - 19)^{40}$ *$40(8x-19)^{39}$*

$320(8x-19)^{39}$

9. $f'(x)$ if $f(x) = \sin(\cos(x))$ *$f' = \cos$, $g' = -\sin x$*

$\cos(\cos(x)) \cdot -\sin(x)$

10. $\frac{d}{dx} (f(x) \cdot g(x))$ in terms of $f(x)$ and $g(x)$

$f'(x) \cdot g(x) + g'(x) \cdot f(x)$